PhishGuard

Project Documentation

Real-Time Browser-Based Phishing Detection and Website Security Analysis

1. Abstract

Phishing attacks use deceptive websites and social-engineering techniques to persuade users to disclose credentials and other sensitive information. PhishGuard is a Chrome-first browser extension developed as a cybersecurity project to analyze webpages in real time and identify multiple indicators associated with suspicious or phishing websites. The system combines URL analysis, IP-address detection, lookalike-domain detection, trusted-domain comparison, credential-form analysis, suspicious-content analysis, and redirect analysis. These findings are processed by a risk engine that produces a score between 0 and 100 and classifies the page as LOW, SUSPICIOUS, HIGH, or CRITICAL.

2. Objectives

Develop a browser extension for real-time webpage security analysis; detect insecure HTTP pages and IP-based access; identify lookalike domains; analyze credential forms; detect suspicious social-engineering language; analyze redirects; produce an explainable risk score; warn users about HIGH and CRITICAL pages; store scan history; and provide configurable security settings.

3. Proposed System

PhishGuard is implemented as a Manifest V3 browser extension. The content scanner coordinates URL/domain, form, page-content, and redirect analysis. The risk engine combines these findings and classifies the page as LOW, SUSPICIOUS, HIGH, or CRITICAL. Results are presented through the popup dashboard and, for HIGH or CRITICAL pages, an in-page warning interface.

4. Main Modules

URL Analyzer: checks HTTP usage, IP addresses, and URL parsing. Domain Analyzer: normalizes domains, extracts base domains, checks trusted domains, and uses Levenshtein similarity for potential lookalikes. Form Analyzer: identifies password forms and checks insecure or cross-origin submission. Page Analyzer: detects selected social-engineering phrases. Redirect Analyzer: uses Chrome navigation information to identify redirects. Risk Engine: combines findings into a capped 0–100 score.

5. Risk Classification

Current classification: LOW 0–29; SUSPICIOUS 30–59; HIGH 60–79; CRITICAL 80–100. The score is capped at 100.

6. Integrated Test Result

A controlled local test page combined HTTP access, IP-based access, an HTTP credential form, and five suspicious social-engineering phrases. The integrated analysis produced a 90/100 CRITICAL result. The system also displayed the individual reasons contributing to the result.

7. Testing

Controlled tests covered HTTP detection, IP detection, lookalike domains, password forms, insecure credential submission, suspicious language, redirect detection, warning configuration, popup live results, scan history, statistics, and the options page. The implemented tests passed for the listed features.

8. Limitations

PhishGuard is a heuristic prototype. The trusted-brand dataset is limited, heuristics can produce false positives, suspicious language does not prove malicious intent, HTTPS does not guarantee legitimacy, sophisticated phishing pages may evade simple rules, Unicode homograph detection can be expanded, and no external threat-intelligence service is currently integrated.

9. Future Scope

Potential improvements include machine-learning-based classification, larger threat-intelligence datasets, Unicode homograph detection, visual similarity analysis, QR-code phishing detection, advanced iframe analysis, improved dynamic-page monitoring, external threat-intelligence APIs, Firefox compatibility, and improved explainability.

10. Conclusion

PhishGuard demonstrates a practical browser-based phishing detection workflow that combines several explainable security indicators. The project integrates URL, domain, form, page-content, and redirect analysis with risk scoring, warnings, settings, and scan history. Controlled testing demonstrated that multiple indicators can be combined into an interpretable CRITICAL result.

Risk Classification Table

Score	Level
0–29	LOW
30–59	SUSPICIOUS
60–79	HIGH
80–100	CRITICAL